
Sharp Mean Data Selection with Two Points

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Abstract

Suppose a learner must approximate the full-data mean by the unweighted average of two selected examples, with repetition allowed. The excess squared loss is exactly the squared error of the selected mean, so the worst relative loss is a normalized geometric rounding constant. A recent open problem determined this constant in one dimension and, by a computational planar argument, when the dimension and selection budget both equal two. We determine it for every dimension. If $F(d, 2)$ is the worst ratio between selected and optimal loss in $\mathbb{R}^d$, then

$$F(d, 2) = 1 + \max \left\{ \frac{1}{3}, \frac{d-1}{2d} \right\}.$$

Thus the one-dimensional outlier construction remains extremal through dimension three. At dimension four, the extremizer changes to a regular simplex. The upper bound follows from a short positive-matrix argument after a variance-nonincreasing Carathéodory reduction. We also give an exact semidefinite characterization for every dimension and budget. Finally, an explicit asymmetric simplex proves $F(4, 3) > 6/5$, exceeding both the one-dimensional and regular-simplex lower bounds. This shows why the general phase diagram is more intricate than the solved two-point case.

1 Introduction

Data selection asks how well a fixed learning rule can perform when it is trained on a small subset chosen from a much larger dataset. This differs from ordinary sampling because the subset is optimized after the full dataset is observed. It also differs from a coresset because the selected examples receive equal weights. Even for squared-loss mean estimation, the

sharp tradeoff between dimension and selection budget was left open by Hanneke et al. (2025a,b).

Let $\mathcal{D}$ be a finite multiset in $\mathbb{R}^d$. Its squared loss and optimal loss are

$$L_{\mathcal{D}}(h) = \frac{1}{|\mathcal{D}|} \sum_{z \in \mathcal{D}} \|h - z\|^2, \quad L_{\mathcal{D}}^* = \min_h L_{\mathcal{D}}(h).$$

For a budget n , select $z_1, \dots, z_n \in \mathcal{D}$, with repetition allowed, and train by taking their mean. Define

$$L_{\mathcal{D}}^*(n) = \min_{z_1, \dots, z_n \in \mathcal{D}} L_{\mathcal{D}} \left(\frac{1}{n} \sum_{j=1}^n z_j \right),$$

$$F(d, n) = \sup_{\mathcal{D} \subset \mathbb{R}^d} \frac{L_{\mathcal{D}}^*(n)}{L_{\mathcal{D}}^*}.$$

The ratio is defined as one when both losses vanish. Translation and the bias-variance identity give

$$\frac{L_{\mathcal{D}}^*(n)}{L_{\mathcal{D}}^*} = 1 + \frac{\min_{z_1, \dots, z_n \in \mathcal{D}} \left\| \frac{1}{n} \sum_{j=1}^n z_j - \mu_{\mathcal{D}} \right\|^2}{\frac{1}{|\mathcal{D}|} \sum_{z \in \mathcal{D}} \|z - \mu_{\mathcal{D}}\|^2}. \quad (1)$$

Hanneke et al. (2025a) proved $F(1, n) = 2n/(2n-1)$ and the dimension-free bounds

$$\frac{2n}{2n-1} \leq F(d, n) \leq \frac{n+1}{n}. \quad (2)$$

Their open-problem note reported $F(2, 2) = 4/3$ through a computational geometric derivation, while all other higher-dimensional entries were open (Hanneke et al., 2025b). We solve the full row $n = 2$.

Theorem 1 (Exact two-point selection). *For every integer $d \geq 1$,*

$$F(d, 2) = 1 + \max \left\{ \frac{1}{3}, \frac{d-1}{2d} \right\}. \quad (3)$$

Equivalently, $F(d, 2) = 4/3$ for $d \leq 3$, while $F(d, 2) = (3d-1)/(2d)$ for $d \geq 4$.

The result has a sharp extremizer transition. A four-point one-dimensional outlier dataset attains $4/3$. It remains worst possible in dimensions two and three. Starting in dimension four, the uniform distribution on a regular d -simplex is worse, and its best pair consists of two distinct vertices.

The proof is elementary once the right matrix is introduced. A linear-programming version of Carathéodory's theorem reduces the centered dataset to at most $d+1$ support points without increasing variance. If every pair average were too large, a shifted Gram matrix would be entrywise positive. Its product with the barycentric weight vector is explicit. In low dimension, positivity forces every weight below $1/4$, which is impossible. In high dimension, the same inequality combines with Jensen's inequality to give the contradiction $1/d > 1/d$.

We also clarify what remains beyond two points. For any fixed barycentric weights, the exact worst geometry is a semidefinite program. A symmetric family with one distinguished vertex reduces this program to a one-dimensional envelope. At $(d, n) = (4, 3)$, a rational member of this family gives

$$F(4, 3) \geq 1 + \frac{529}{2592} > \frac{6}{5}. \quad (4)$$

The right side of the strict inequality is the one-dimensional lower bound. The regular-simplex excess is only $1/6$. Hence neither familiar construction describes the general answer.

Relation to approximate Carathéodory. Approximate Carathéodory results replace a convex combination by an equal-weight average and usually normalize by a uniform radius (Barman, 2018; Mirrokni et al., 2017). Tinarrage (2026) recently found the sharp dimension-dependent Euclidean radius for simplex grids. Our denominator in (1) is instead the variance around the represented point. It can vanish as that point approaches a face. This point-dependent normalization produces the outlier extremizer and the transition in (3). Regular-grid rounding in barycentric norms has also been studied in global optimization (Bonze et al., 2014), but that geometry does not optimize over the data Gram matrix or normalize by its variance.

2 Reduction to a Small Centered Support

We use the following direct form of the reduction behind the mean-estimation argument of Hanneke et al. (2025a).

Lemma 2 (Variance-nonincreasing reduction). *Let X be a finitely supported random vector in $\mathbb{R}^d$. There is a random vector Y , supported on at most $d+1$ points from $\text{supp}(X)$, such that $\mathbb{E}Y = \mathbb{E}X$ and*

$$\mathbb{E}\|Y - \mathbb{E}Y\|^2 \leq \mathbb{E}\|X - \mathbb{E}X\|^2.$$

Proof. After translation, assume $\mathbb{E}X = 0$. Over weights $w_z \geq 0$ on the support of X , minimize $\sum_z w_z \|z\|^2$ subject to $\sum_z w_z = 1$ and $\sum_z w_z z = 0$. The law of X is feasible. A basic optimal solution has at most $d+1$ positive entries because there are at most $d+1$ independent equality constraints. The associated Y has the required mean and no larger second moment. $\square$

Consequently, to upper-bound (1), it suffices to consider points $x_1, \dots, x_m$ with positive weights $p_1, \dots, p_m$, where $m \leq d+1$, and normalize them so that

$$\sum_{i=1}^m p_i x_i = 0, \quad \sum_{i=1}^m p_i \|x_i\|^2 = 1. \quad (5)$$

If two support points have average of squared norm at most β , the same two points belong to the original dataset and give an excess ratio at most β .

Let $r_i = \|x_i\|^2$ and $g_{ij} = \langle x_i, x_j \rangle$. We will use

$$\sum_j p_j g_{ij} = 0, \quad \sum_j p_j r_j = 1. \quad (6)$$

The reduction also gives an exact geometric formulation that isolates the difference from ordinary approximate Carathéodory bounds. Let

$$\mathcal{A}_{m,n} = \left\{ q = k/n : k \in \mathbb{N}^m, \sum_i k_i = n \right\}.$$

Proposition 3 (Normalized spherical-simplex covering). *For every $d, n \geq 1$,*

$$F(d, n) - 1 = \sup_{q \in \mathcal{A}_{m,n}} \frac{\min_{y \in \mathcal{A}_{m,n}} \|Uq - y\|^2}{1 - \|y\|^2}, \quad (7)$$

where the supremum ranges over $2 \leq m \leq d+1$, unit vectors $u_1, \dots, u_m \in \mathbb{S}^{d-1}$, and $p \in \Delta_m$. Here U has columns u_i , $y = Up = \sum_i p_i u_i$, and $\|y\| < 1$. Zero weights may be removed.

Proof. In the linear program from Lemma 2, dual complementary slackness gives a vector v and scalar α such that every selected support point satisfies

$$\|x_i\|^2 = \alpha + \langle v, x_i \rangle.$$

Thus the selected points lie on a common sphere centered at $v/2$. Write $x_i = Ru_i + v/2$ with $\|u_i\| = 1$. Since $\sum_i p_i x_i = 0$, the represented point satisfies $y = \sum_i p_i u_i = -v/(2R)$. Therefore

$$\begin{aligned} \sum_i p_i \|x_i\|^2 &= R^2(1 - \|y\|^2), \\ \left\| \sum_i q_i x_i \right\|^2 &= R^2 \|Uq - y\|^2. \end{aligned}$$

Their ratio is (7). Conversely, any spherical configuration in the display produces centered points $x_i = u_i - y$ with variance $1 - \|y\|^2$. Real weights are limits of rational finite-multiset weights. $\square$

For a simplex in the unit ball, Tinarrage (2026) bounds the numerator in (7) uniformly by a sharp dimension-dependent radius. Equation (7) divides by $1 - \|y\|^2$. This denominator approaches zero near the spherical boundary, so a uniform covering theorem cannot determine $F(d, n)$. Theorem 1 shows that at $n = 2$ the competition between the central regular simplex and this boundary regime is exact.

3 Proof of the Sharp Constant

Set

$$\begin{aligned} \beta_d &= \max \left\{ \frac{1}{3}, \frac{d-1}{2d} \right\}, \\ \gamma_d &= 1 - 2\beta_d = \min \left\{ \frac{1}{3}, \frac{1}{d} \right\}. \end{aligned} \quad (8)$$

Lemma 4 (Positive-matrix pair bound). *Every system satisfying (5) with $m \leq d+1$ contains indices i, j , possibly equal, such that*

$$\left\| \frac{x_i + x_j}{2} \right\|^2 \leq \beta_d.$$

Proof. Suppose for contradiction that every pair, including repetitions, has squared average strictly larger than β_d . Averaging the inequality for $(x_i + x_j)/2$ over j with weights p_j and using (6) gives

$$\frac{r_i + 1}{4} > \beta_d.$$

Hence $r_i > 1 - 2\gamma_d$. Define

$$\ell_i = r_i - (1 - 2\gamma_d) > 0, \quad h_i = \frac{\ell_i}{2}.$$

The normalization gives $\sum_i p_i \ell_i = 2\gamma_d$. The pairwise assumption is equivalent to

$$g_{ij} > \gamma_d - \frac{\ell_i + \ell_j}{2}.$$

Therefore the symmetric matrix

$$B_{ij} = g_{ij} - \gamma_d + \frac{\ell_i + \ell_j}{2} \quad (9)$$

is entrywise positive. Centering and normalization yield

$$(Bp)_i = h_i, \quad p^\top Bp = \gamma_d, \quad B_{ii} = 1 - 3\gamma_d + 4h_i. \quad (10)$$

Write $a = 1 - 3\gamma_d \geq 0$. Since every off-diagonal entry is positive,

$$h_i = \sum_j p_j B_{ij} > p_i B_{ii} = p_i(a + 4h_i). \quad (11)$$

If $d \leq 3$, then $\gamma_d = 1/3$ and $a = 0$. Equation (11) gives $p_i < 1/4$ for every i . This contradicts $\sum_i p_i = 1$ because $m \leq d+1 \leq 4$.

Now suppose $d \geq 4$. Then $\gamma_d = 1/d$ and $a = (d-3)/d > 0$. Equation (11) again gives $p_i < 1/4$, followed by

$$h_i > \frac{ap_i}{1 - 4p_i}.$$

Multiply by p_i and sum. Since $\sum_i p_i h_i = \gamma_d$,

$$\gamma_d > a \sum_{i=1}^m \frac{p_i^2}{1 - 4p_i}. \quad (12)$$

The function $f(x) = x^2/(1-4x)$ satisfies $f''(x) = 2/(1-4x)^3 > 0$ on $[0, 1/4)$. Jensen's inequality and $m \leq d+1$ imply

$$\sum_{i=1}^m \frac{p_i^2}{1 - 4p_i} \geq mf(1/m) = \frac{1}{m-4} \geq \frac{1}{d-3}.$$

Substitution into (12) gives

$$\frac{1}{d} = \gamma_d > \frac{d-3}{d} \frac{1}{d-3} = \frac{1}{d},$$

which is impossible. $\square$

Proof of Theorem 1. Lemma 2 and Lemma 4 give the upper bound $F(d, 2) \leq 1 + \beta_d$.

For the first lower bound, take a one-dimensional dataset containing three copies of zero and one copy of 1. Its mean is $1/4$, its variance is $3/16$, and every two-point average is in $\{0, 1/2, 1\}$. The closest selected mean has squared error $1/16$, so its excess ratio is $1/3$. This construction embeds in every $\mathbb{R}^d$.

For the second lower bound, let $v_0, \dots, v_d$ be the unit vertices of a regular d -simplex, so that $\sum_i v_i = 0$ and $\langle v_i, v_j \rangle = -1/d$ for $i \neq j$. The variance is one. Repeating a vertex gives squared norm one, while two distinct vertices give

$$\left\| \frac{v_i + v_j}{2} \right\|^2 = \frac{1}{4} \left(2 - \frac{2}{d} \right) = \frac{d-1}{2d}.$$

Taking the larger of the two lower bounds completes the proof. $\square$

The high-dimensional extremizer is rigid after support reduction.

Corollary 5 (Equality for $d \geq 4$). *Let $d \geq 4$. Suppose a centered, variance-one system with positive weights, supported on at most $d + 1$ points, attains the upper bound in Lemma 4. Then it has exactly $d + 1$ points, all weights equal $1/(d + 1)$, and its Gram matrix is that of the unit regular d -simplex.*

Proof. Repeat the proof of Lemma 4 with non-strict inequalities. The matrix B is entrywise nonnegative and

$$h_i \geq p_i(a + 4h_i), \quad a = \frac{d-3}{d} > 0.$$

Hence $p_i < 1/4$ and

$$\frac{1}{d} = \sum_i p_i h_i \geq a \sum_i \frac{p_i^2}{1 - 4p_i} \geq \frac{a}{m-4} \geq \frac{a}{d-3} = \frac{1}{d}.$$

Every inequality is therefore an equality. Equality in Jensen's inequality gives $p_i = 1/m$, while equality in the support-size step gives $m = d + 1$. It follows that $h_i = 1/d$, so $\ell_i = 2/d$ and $r_i = 1$. Equality in $h_i = \sum_j p_j B_{ij} \geq p_i B_{ii}$ forces $B_{ij} = 0$ for $i \neq j$. Equation (9) then gives $g_{ij} = -1/d$. This is exactly the regular-simplex Gram matrix. $\square$

The strict contradiction proves the non-strict upper bound by negating the claim that no qualifying pair exists. No compactness or limiting argument is needed after the finite-support reduction.

4 The General Budget as an SDP

The proof above is special to pairs, but the reduced problem has an exact finite-dimensional representation for every n . Let

$$\mathcal{K}_{m,n} = \left\{ k \in \mathbb{N}^m : \sum_{i=1}^m k_i = n \right\}.$$

Proposition 6 (Fixed-weight geometry program). *For p in the relative interior of the probability simplex, define*

$$C_{m,n}(p) = \max_{G,t} t \quad \text{subject to} \quad G \succeq 0, \quad Gp = 0, \quad (13)$$

$$\sum_i p_i G_{ii} = 1, \quad \frac{k^\top G k}{n^2} \geq t \quad (k \in \mathcal{K}_{m,n}). \quad (14)$$

Then

$$F(d, n) - 1 = \sup_{2 \leq m \leq d+1} \sup_p C_{m,n}(p). \quad (15)$$

Proof. For reduced points $x_1, \dots, x_m$, let G be their Gram matrix. Conditions (5) are exactly $G \succeq 0$, $Gp = 0$, and $\sum_i p_i G_{ii} = 1$. A count vector k produces squared selected mean $k^\top G k / n^2$. Conversely, every feasible G is the Gram matrix of vectors in dimension $\text{rank}(G) \leq m - 1 \leq d$. Taking the best count vector, then the worst geometry, weights, and support size gives (15). Arbitrary real weights follow by approximation with finite multisets. $\square$

For fixed p , (13) is an ordinary semidefinite program with $\binom{n+m-1}{m-1}$ linear inequalities. The outer maximization over p is the genuine difficulty. The following construction shows that uniform weights and two-point supports do not suffice.

Proposition 7 (One-distinguished-vertex family). *Fix $m \geq 3$, $n \geq 2$, $a \in (0, 1)$, and $s \in [0, 1]$. Let $Q_r(t)$ be the minimum of $\sum_{i=1}^r k_i^2$ over nonnegative integers summing to t . For $0 \leq l \leq n$, set*

$$E_l(a, s) = s \frac{(l - an)^2}{n^2 a (1 - a)} + (1 - s) \frac{(m-1)Q_{m-1}(n-l) - (n-l)^2}{(m-2)n^2(1-a)}. \quad (16)$$

For every $d \geq m - 1$,

$$F(d, n) - 1 \geq \sup_{a \in (0,1)} \min_{s \in [0,1]} \min_{0 \leq l \leq n} E_l(a, s).$$

In particular, (4) holds.

Proof. Use barycentric weights $(a, b, \dots, b)$, where $b = (1 - a)/(m - 1)$. Put a fraction s of the variance on an axis separating the distinguished vertex from the other vertices. Put the remaining fraction on a regular $(m - 2)$ -simplex carried by the opposite face. If the distinguished point is selected l times, the axial squared error is the first term of (16). Balancing the remaining $n - l$ counts across the opposite face minimizes their sum of squares and gives the second term. Full coordinates and the variance calculation are in the supplement.

For (4), take $m = 5$, $n = 3$, $a = 2/5$, and $s = 3/16$. Direct substitution gives

$$(E_0, E_1, E_2, E_3) = \left(\frac{119}{432}, \frac{529}{2592}, \frac{89}{432}, \frac{9}{32} \right).$$

Their minimum is $529/2592$. Finally, $529/2592 - 1/5 = 53/12960 > 0$. $\square$

5 Discussion

Theorem 1 identifies an exact dimensional transition that neither the dimension-free sampling bound nor

the unnormalized simplex covering radius reveals. The proof also explains the threshold. Entrywise positivity imposes $p_i < 1/4$. Up to dimension three, support reduction makes this impossible. From dimension four onward, the convex barrier $x^2/(1-4x)$ produces the regular-simplex constant.

The remaining values $F(d, n)$ contain additional regimes. Proposition 7 already places an asymmetric full-dimensional simplex above both elementary lower bounds at $(4, 3)$. The SDP in Proposition 6 offers a precise route to the general problem, but a complete solution must control the outer weight optimization rather than only solve a Euclidean grid-covering problem. A promising interpretation is minimax rounding of barycentric weights under an adversarial Gram metric. The two-point proof gives a closed-form dual certificate for that game. Finding its higher-budget analogue is the main open question.

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A Exact Loss Reduction

Let $\mu_{\mathcal{D}} = |\mathcal{D}|^{-1} \sum_{z \in \mathcal{D}} z$. Expanding around the mean gives, for every $h \in \mathbb{R}^d$,

$$L_{\mathcal{D}}(h) = \frac{1}{|\mathcal{D}|} \sum_{z \in \mathcal{D}} \|z - \mu_{\mathcal{D}}\|^2 + \|h - \mu_{\mathcal{D}}\|^2.$$

The cross term vanishes. Therefore $L_{\mathcal{D}}^*$ is the empirical variance and the selected-mean excess is exactly its squared distance to $\mu_{\mathcal{D}}$. This proves equation (1) in the main paper.

For completeness, the linear program in Lemma 1 has one normalization equation and at most d independent mean equations. Every linear program with a finite feasible set of variables has an optimal basic feasible solution. The number of its positive coordinates is at most the rank of the equality system, hence at most $d + 1$. If the support lies in a proper affine subspace, the rank and support bound only decrease.

B Details for the One-Distinguished Family

We construct the points used in the one-distinguished-vertex proposition of the main paper. Let $w_1, \dots, w_{m-1}$ be unit vertices of a regular $(m-2)$ -simplex in $\mathbb{R}^{m-2}$, so that

$$\sum_{i=1}^{m-1} w_i = 0, \quad \langle w_i, w_j \rangle = -\frac{1}{m-2} \quad (i \neq j).$$

Use an orthogonal axial coordinate and define

$$\begin{aligned} x_0 &= \sqrt{s} \sqrt{\frac{1-a}{a}} e, \\ x_i &= -\sqrt{\frac{sa}{1-a}} e + \sqrt{\frac{1-s}{1-a}} w_i, \\ 1 \leq i \leq m-1. \end{aligned}$$

Give x_0 weight a and every other point weight $b = (1-a)/(m-1)$. The weighted axial mean is

$$a \sqrt{\frac{1-a}{a}} - (1-a) \sqrt{\frac{a}{1-a}} = 0,$$

and the contrast mean vanishes by regular-simplex symmetry. The weighted axial second moment is s , while the weighted contrast second moment is $1-s$. Thus the system is centered and has variance one.

Suppose a count vector selects x_0 exactly l times and selects the other points with counts $k_1, \dots, k_{m-1}$. The axial component of the selected mean equals

$$\frac{\sqrt{s}}{n} \left(l \sqrt{\frac{1-a}{a}} - (n-l) \sqrt{\frac{a}{1-a}} \right) = \sqrt{s} \frac{l - an}{n \sqrt{a(1-a)}}.$$

For $r = m - 1$ and $t = n - l$, regular-simplex identities give

$$\left\| \sum_{i=1}^r k_i w_i \right\|^2 = \frac{r \sum_i k_i^2 - t^2}{r - 1}.$$

The minimum of $\sum_i k_i^2$ is obtained by balancing the integer counts. If $t = ru + v$ with $0 \leq v < r$, then

$$Q_r(t) = (r - v)u^2 + v(u + 1)^2.$$

Combining the orthogonal axial and contrast terms gives equation (15) in the main paper.

For the rational example $m = 5$, $n = 3$, $a = 2/5$, and $s = 3/16$, the balanced sums are $Q_4(3) = 3$, $Q_4(2) = 2$, $Q_4(1) = 1$, and $Q_4(0) = 0$. Substitution gives

$$\begin{aligned} E_0 &= \frac{3}{16} \frac{2}{3} + \frac{13}{16} \frac{5}{27} = \frac{119}{432}, \\ E_1 &= \frac{3}{16} \frac{1}{54} + \frac{13}{16} \frac{20}{81} = \frac{529}{2592}, \\ E_2 &= \frac{3}{16} \frac{8}{27} + \frac{13}{16} \frac{5}{27} = \frac{89}{432}, \\ E_3 &= \frac{3}{16} \frac{3}{2} = \frac{9}{32}. \end{aligned}$$

C Dual View of the Fixed-Weight SDP

This section records a useful equivalent interpretation of the fixed-weight SDP in the main paper. Let

$$\mathcal{A}_{m,n} = \left\{ q = k/n : k \in \mathbb{N}^m, \sum_i k_i = n \right\}.$$

For a feasible Gram matrix, $Gp = 0$ implies $q^\top Gq = (q - p)^\top G(q - p)$. By minimax duality for the finite family of grid points, the fixed-weight value can be written as

$$M_\lambda = \sum_{q \in \mathcal{A}_{m,n}} \lambda_q (q - p)(q - p)^\top,$$

$$\begin{aligned} C_{m,n}(p) &= \min_{\lambda \in \Delta(\mathcal{A}_{m,n})} \max_{G \succeq 0, Gp=0} \text{tr}(GM_\lambda) \\ &\quad \text{subject to } \text{tr}(\text{diag}(p)G) = 1. \end{aligned}$$

The inner maximum is the largest generalized eigenvalue of the grid-rounding second moment relative to $\text{diag}(p)$ on the subspace $p^\perp$. Thus the problem asks for a distribution over denominator- n rounding points whose worst covariance direction is as small as possible. The distribution need not be unbiased. This last point is essential near a face of the simplex, where biased rounding toward the face is better than enforcing the original barycentric mean.

For $n = 2$, Lemma 2 constructs a dual obstruction indirectly. If all grid points had value above β_d , the

shifted matrix B in equation (8) would be entrywise positive. Equations (9) and (10) then contradict the support-size constraint. This is why the pair problem admits a dimension-explicit certificate without solving the outer optimization over p .

D Checks on the Extremal Constructions

For the outlier dataset $\{0, 0, 0, 1\}$, the empirical mean is $1/4$ and

$$L_{\mathcal{D}}^* = \frac{1}{4} \left(3 \left(\frac{1}{4} \right)^2 + \left(\frac{3}{4} \right)^2 \right) = \frac{3}{16}.$$

The best two-point empirical mean is either 0 or $1/2$, both at distance $1/4$ from the full mean. Hence

$$\frac{L_{\mathcal{D}}^*(2)}{L_{\mathcal{D}}^*} = 1 + \frac{1/16}{3/16} = \frac{4}{3}.$$

For a regular d -simplex with unit vertices, the mean is zero and the variance is one. A repeated vertex has squared norm one. A distinct pair has squared mean norm $(d-1)/(2d)$. This is no larger than one for every $d \geq 1$, so the best pair is distinct. The resulting ratio is $(3d-1)/(2d)$.

The two lower bounds agree when $(d-1)/(2d) = 1/3$, which occurs at $d = 3$. The regular-simplex bound is strictly larger for $d \geq 4$.